

Henry Lee

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EDUCATION

Purdue University | West Lafayette, IN

GPA: 3.5/4.0

Bachelor of Science, Major in Computer Science, Minor in Mathematics

August 2023 – Present

- **Relevant Coursework:** Object-Oriented Programming, Computer Architecture, Data Structures & Algorithms, Systems Programming, Algorithms Analysis, Intro to AI, Data Mining & Machine Learning, Intro to Robotics

EXPERIENCE

Software Engineering Intern

June 2026 – August 2026

Astronautics Corporation of America

Oak Creek, WI

- Enhanced an Airbus ground-tool web application by implementing unique identifiers for UI elements
- Built Cypress test suites to validate core application functionality

Machine Learning Co-Op

August 2025 – December 2025

Purdue University: The Data Mine (Empire Cooler)

West Lafayette, IN

- Developed a Python retrieval tool and data pipelines for extracting repair information from technical datasets
- Worked in an Agile team to design, prototype, and refine AI-powered solutions

Software Engineering Intern

October 2024 – August 2025

Echopath LLC

Indianapolis, IN (Remote)

- Built Python, Rust, and SQL tools to automate financial and ticketing workflows
- Enhanced SQL database performance for 700K+ records through indexing and schema normalization

Software Engineering Co-Op

August 2024 – December 2024

Purdue University: The Data Mine (Concrete Engine)

West Lafayette, IN

- Collaborated in a startup environment to develop software for AI/HPC workflows
- Developed MongoDB models and researched scalable cloud-storage and authentication solutions

Data Science Co-Op

January 2024 – May 2024

Purdue University: The Data Mine (Wikimedia Deutschland)

West Lafayette, IN

- Detected and resolved 900+ data inconsistencies to improve dataset quality and reliability
- Constructed Python and SPARQL pipelines to retrieve and validate external data sources

PROJECTS

Shape Synthesis | Python, PyTorch, Open3D, Machine Learning

March 2026 – May 2026

- Developed a machine learning model for 3D shape synthesis from geometric data
- Implemented alpha-shape surface reconstruction to generate watertight meshes from point clouds

Custom Unix Shell | C/C++, Linux, Flex & Bison

March 2026

- Built a Unix-like shell supporting piping, I/O redirection, background processes, and signals
- Designed a parser for pipelines, redirection, and concurrent command execution

Custom Memory Allocator | C, Linux, Bash, Systems Programming

January 2026 - February 2026

- Implemented a custom malloc/free allocator using boundary tags and segregated free lists
- Optimized realloc and heap coalescing to improve memory efficiency and reduce copying overhead

Ticket Search Utility | Python, Rust, Flask, SQL, APIs

April 2025 – August 2025

- Developed a scalable ticket-search platform supporting 700K+ support records
- Integrated ConnectWise APIs and optimized SQL queries to significantly reduce search latency

SimpleC Compiler | Lex, Yacc, C, x86-64 Assembly, Makefile

October 2024 - December 2024

- Designed and implemented a one-pass compiler targeting x86-64 assembly
- Generated assembly code for control flow, expressions, function calls, and memory management

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Rust, JavaScript, TypeScript, SQL, MATLAB, C#, Linux, Bash

Libraries/Frameworks: React, Next.js, Flask, Node.js, scikit-learn, Open3D, Pandas, Matplotlib, JUnit, Cypress

Developer Tools: Git, Linux, VS Code, PyCharm, IntelliJ, Google Colab, MongoDB